

Midterm 1
Friday, February 25, 2022

Name

- 6 problems total
- 75 minutes (4:00-5:15)
- Only writing instruments and nongraphing calculators are allowed
- Harder / longer problems are in the back. Allot time appropriately
- You can tear away the formula sheets in the back. You don't need to turn them in with your test; if you do tear them off, make sure to recycle them when you leave
- You can use the backs of pages if you need extra space for calculations
- If you feel you are missing any formulas, let me know

Good luck!

Problem 1:

a) Calculate curl and divergence of the following vector field. State whether the field is conservative, solenoidal, both, or neither.

$$\mathbf{A} = \hat{\mathbf{R}} 5 \exp(-10R^2) / R^2$$

b) Calculate curl and divergence of the following vector field. State whether the field is conservative, solenoidal, both, or neither.

$$\mathbf{A} = \hat{\mathbf{x}}(2x + 3yz) + \hat{\mathbf{y}}(2y + 3xz) + \hat{\mathbf{z}}(2z + 3xy)$$

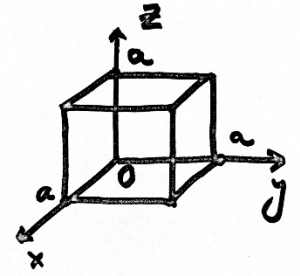
c) Some vector field $\mathbf{A}$ is related to a scalar field f as $\mathbf{A} = \nabla f$. What is the value of $\nabla \times \mathbf{A}$? Explain briefly (1-2 sentences).

Problem 2: (*Hint: If you know the material well, this problem will take very little time to do.*)

a) Calculate the flux of the current-density vector field

$$\mathbf{J} = \hat{\mathbf{x}} (3x - 2y) + \hat{\mathbf{y}} (x - 3y) \text{ [A/m}^2\text{]}$$

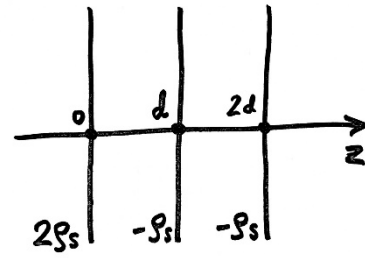
through a cube of side a in the first octant of the Cartesian coordinate system, with one corner at the coordinate origin and three edges aligned with the coordinate axes, as in the picture on the right.



b) Can $\mathbf{J}$ be the current density in the static (time-independent) case? Explain briefly (1-2 sentences).

c) Will the total amount of charge inside this cube change with time? Explain briefly (1-2 sentences).

Problem 3: Three infinite parallel planes have surface charge densities $2\rho_s$, $-\rho_s$, $-\rho_s$ ($\rho_s > 0$). They are normal to the z -axis and placed at $z=0$, $z=d$, and $z=2d$, respectively, as in the figure. (*Read carefully. This is NOT the same as the practice midterm!*). Assume the planes are in vacuum (permittivity ϵ_0).



a) Find $\mathbf{E}$ everywhere in space. Graph $E(z)$ versus z .

b) Calculate and graph the electric potential $V(z)$. Assume $V(0)=0$.

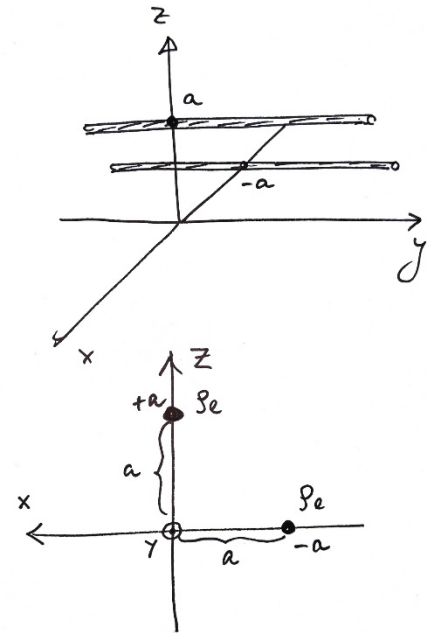
Problem 4: As discussed in class and in homework, the potential from an infinite thin wire with line charge density ρ_l , which is aligned with the z -axis, can be written as

$$V(r) = \frac{\rho_l}{2\pi\epsilon} \ln\left(\frac{r_{ref}}{r}\right)$$

where r_{ref} is normal distance to the reference point.

Generalize this result and use the principle of superposition to obtain the potential arising from two infinitely long wires, both parallel to the y -axis, one located at $x = -a$ and $z = 0$, the other at $x = 0$ and $z = a$ (see picture). The two wires have the same line charge density ρ_l . The material throughout is vacuum.

Write down the potential at a point of observation P with arbitrary (x, y, z) coordinates. The reference point is the coordinate origin $(0,0,0)$.



Problem 5: An infinitely long cylinder of radius a is charged with a uniform volume charge density ρ_{v0} . There is no charge outside the cylinder. Assume the permittivity of the cylinder is $\epsilon_r \epsilon_0$ ($\epsilon_r > 1$), whereas that of the surrounding medium is ϵ_0 .

a) Use Gauss's law to calculate $\mathbf{D}$ everywhere in space. Graph $D(r)$ vs r .

b) Calculate and graph $E(r)$ vs r everywhere. Calculate and graph $P(r)$ vs r everywhere.

c) Calculate and graph the electric potential $V(r)$, assuming it is zero on the cylinder axis.

Problem 6:

An infinitely thin spherical shell of radius a has a uniform surface charge density $\rho_s > 0$. Assume the surrounding medium is air (permittivity ϵ_0).

a) Use Gauss's law to calculate the electric field $\mathbf{E}$ everywhere in space. Graph $E(R)$ vs R .

b) Another thin spherical shell, with the surface charge density $-\rho_s/4$ and radius $b=2a$, is placed concentrically around the shell from part a). Based on your result from a) and using the principle of superposition, calculate and graph the total electric field $E(R)$ vs R everywhere in space for the two spheres together.

c) Calculate the potential difference between the inner and outer shells, i.e., $V(a)-V(b)$.

Mathematical and Physical Models

Maxwell's Equations for Electrostatics

Name	Differential Form	Integral Form
Gauss's law	$\nabla \cdot \mathbf{D} = \rho_v$	$\oint_S \mathbf{D} \cdot d\mathbf{s} = Q$
Kirchhoff's law	$\nabla \times \mathbf{E} = 0$	$\oint_C \mathbf{E} \cdot d\mathbf{l} = 0$

Electric Field

Current density	$\mathbf{J} = \rho_v \mathbf{u}$	Point charge	$\mathbf{E} = \hat{\mathbf{R}} \frac{q}{4\pi\epsilon R^2}$
Poisson's equation	$\nabla^2 V = -\frac{\rho_v}{\epsilon}$	Many point charges	$\mathbf{E} = \frac{1}{4\pi\epsilon} \sum_{i=1}^N \frac{q_i (\mathbf{R} - \mathbf{R}_i)}{ \mathbf{R} - \mathbf{R}_i ^3}$
Laplace's equation	$\nabla^2 V = 0$	Volume distribution	$\mathbf{E} = \frac{1}{4\pi\epsilon} \int_{V'} \hat{\mathbf{R}}' \frac{\rho_v dV'}{R'^2}$
Resistance	$R = \frac{l}{\int_S \sigma \mathbf{E} \cdot d\mathbf{s}}$	Surface distribution	$\mathbf{E} = \frac{1}{4\pi\epsilon} \int_{S'} \hat{\mathbf{R}}' \frac{\rho_s ds'}{R'^2}$
Boundary conditions	Table 4-3	Line distribution	$\mathbf{E} = \frac{1}{4\pi\epsilon} \int_{l'} \hat{\mathbf{R}}' \frac{\rho_l dl'}{R'^2}$
Capacitance	$C = \frac{Q}{-\int \mathbf{E} \cdot d\mathbf{l}}$	Infinite sheet of charge	$\mathbf{E} = \hat{\mathbf{z}} \frac{\rho_s}{2\epsilon_0}$
RC relation	$RC = \frac{\epsilon}{\sigma}$	Infinite line of charge	$\mathbf{E} = \frac{\mathbf{D}}{\epsilon_0} = \hat{\mathbf{r}} \frac{D_r}{\epsilon_0} = \hat{\mathbf{r}} \frac{\rho_l}{2\pi\epsilon_0 r}$
Energy density	$w_e = \frac{1}{2} \epsilon E^2$	Dipole	$\mathbf{E} = \frac{qd}{4\pi\epsilon_0 R^3} (\hat{\mathbf{R}} 2 \cos \theta + \hat{\boldsymbol{\theta}} \sin \theta)$
		Relation to V	$\mathbf{E} = -\nabla V$

$$V = \frac{1}{4\pi\epsilon} \int_{V'} \frac{\rho_v}{R'} dV' \quad (\text{volume distribution}), \quad (4.48a)$$

$$V = \frac{1}{4\pi\epsilon} \int_{S'} \frac{\rho_s}{R'} ds' \quad (\text{surface distribution}), \quad (4.48b)$$

$$V = \frac{1}{4\pi\epsilon} \int_{l'} \frac{\rho_l}{R'} dl' \quad (\text{line distribution}). \quad (4.48c)$$

Here, $\mathbf{R}'$ denotes the vector pointing from source at $\mathbf{R}_0$ to point of observation at $\mathbf{R}$, i.e., $\mathbf{R}' = \mathbf{R} - \mathbf{R}_0$

Charge-Current Continuity

$$\nabla \cdot \mathbf{J} = -\frac{\partial \rho_v}{\partial t}$$

Conductor Charge Dissipation

$$\rho_v(t) = \rho_{v0} e^{-(\sigma/\epsilon)t} = \rho_{v0} e^{-t/\tau_c}$$

Current Density

$$\text{Conduction} \quad \mathbf{J}_c = \sigma \mathbf{E}$$

Integral form

$$\oint \vec{J} \cdot d\vec{S} = -\frac{d}{dt} Q_{\text{inside}}$$

$$\begin{array}{l|l} \vec{D} = \epsilon \vec{E} & \vec{D} = \vec{P} + \epsilon_0 \vec{E} \\ \vec{P} = (\epsilon_r - 1) \epsilon_0 \vec{E} = \frac{\epsilon_r - 1}{\epsilon_r} \vec{D} & \vec{J} = \sigma \vec{E} \\ & \nabla \cdot \vec{J} + \frac{\partial \rho_v}{\partial t} = 0 \end{array}$$

$$V(\vec{R}) = - \int_{R_{ref}}^{\vec{R}} \vec{E} \cdot d\vec{l}' \xrightarrow{\text{spherical}} - \int_{R_{ref}}^R E(R) dR_0$$

$\swarrow \text{cylind.}$
 $- \int_{r_{ref}}^r E(r) dr$

$$\oint \vec{D} \cdot d\vec{S} = \int \rho_v(\vec{R}_0) dV_0$$

spherical $4\pi R^2 D(R) = Q_{inside} = 4\pi \int_0^R \rho_v(R_0) R_0^2 dR_0$

cylindrical $2\pi r h D(r) = Q_{inside} = 2\pi h \int_0^r \rho_v(r_0) r_0 dr_0$

Infinite line $\vec{E} = \hat{r} \frac{\rho_l}{2\pi\epsilon r} = \frac{\rho_l}{2\pi\epsilon} \frac{\hat{r}}{r^2}$

Infinite plane $\vec{E} = \frac{\rho_s}{2\epsilon} \begin{cases} \hat{z}, & z > 0 \\ -\hat{z}, & z < 0 \end{cases}$

Coulomb's law $\vec{E} = \frac{1}{4\pi\epsilon} \frac{Q}{R^2} \hat{R} = \frac{1}{4\pi\epsilon} \frac{Q}{R^3} \vec{R}$

Table 3-1 Summary of vector relations.

	Cartesian Coordinates	Cylindrical Coordinates	Spherical Coordinates
Coordinate variables	x, y, z	r, ϕ, z	R, θ, ϕ
Vector representation $\mathbf{A} =$	$\hat{\mathbf{x}}A_x + \hat{\mathbf{y}}A_y + \hat{\mathbf{z}}A_z$	$\hat{\mathbf{r}}A_r + \hat{\boldsymbol{\phi}}A_\phi + \hat{\mathbf{z}}A_z$	$\hat{\mathbf{R}}A_R + \hat{\boldsymbol{\theta}}A_\theta + \hat{\boldsymbol{\phi}}A_\phi$
Magnitude of A $ \mathbf{A} =$	$\sqrt{A_x^2 + A_y^2 + A_z^2}$	$\sqrt{A_r^2 + A_\phi^2 + A_z^2}$	$\sqrt{A_R^2 + A_\theta^2 + A_\phi^2}$
Position vector $\overrightarrow{OP_1} =$	$\hat{\mathbf{x}}x_1 + \hat{\mathbf{y}}y_1 + \hat{\mathbf{z}}z_1,$ for $P(x_1, y_1, z_1)$	$\hat{\mathbf{r}}r_1 + \hat{\mathbf{z}}z_1,$ for $P(r_1, \phi_1, z_1)$	$\hat{\mathbf{R}}R_1,$ for $P(R_1, \theta_1, \phi_1)$
Base vectors properties	$\hat{\mathbf{x}} \cdot \hat{\mathbf{x}} = \hat{\mathbf{y}} \cdot \hat{\mathbf{y}} = \hat{\mathbf{z}} \cdot \hat{\mathbf{z}} = 1$ $\hat{\mathbf{x}} \cdot \hat{\mathbf{y}} = \hat{\mathbf{y}} \cdot \hat{\mathbf{z}} = \hat{\mathbf{z}} \cdot \hat{\mathbf{x}} = 0$ $\hat{\mathbf{x}} \times \hat{\mathbf{y}} = \hat{\mathbf{z}}$ $\hat{\mathbf{y}} \times \hat{\mathbf{z}} = \hat{\mathbf{x}}$ $\hat{\mathbf{z}} \times \hat{\mathbf{x}} = \hat{\mathbf{y}}$	$\hat{\mathbf{r}} \cdot \hat{\mathbf{r}} = \hat{\boldsymbol{\phi}} \cdot \hat{\boldsymbol{\phi}} = \hat{\mathbf{z}} \cdot \hat{\mathbf{z}} = 1$ $\hat{\mathbf{r}} \cdot \hat{\boldsymbol{\phi}} = \hat{\boldsymbol{\phi}} \cdot \hat{\mathbf{z}} = \hat{\mathbf{z}} \cdot \hat{\mathbf{r}} = 0$ $\hat{\mathbf{r}} \times \hat{\boldsymbol{\phi}} = \hat{\mathbf{z}}$ $\hat{\boldsymbol{\phi}} \times \hat{\mathbf{z}} = \hat{\mathbf{r}}$ $\hat{\mathbf{z}} \times \hat{\mathbf{r}} = \hat{\boldsymbol{\phi}}$	$\hat{\mathbf{R}} \cdot \hat{\mathbf{R}} = \hat{\boldsymbol{\theta}} \cdot \hat{\boldsymbol{\theta}} = \hat{\boldsymbol{\phi}} \cdot \hat{\boldsymbol{\phi}} = 1$ $\hat{\mathbf{R}} \cdot \hat{\boldsymbol{\theta}} = \hat{\boldsymbol{\theta}} \cdot \hat{\boldsymbol{\phi}} = \hat{\boldsymbol{\phi}} \cdot \hat{\mathbf{R}} = 0$ $\hat{\mathbf{R}} \times \hat{\boldsymbol{\theta}} = \hat{\boldsymbol{\phi}}$ $\hat{\boldsymbol{\theta}} \times \hat{\boldsymbol{\phi}} = \hat{\mathbf{R}}$ $\hat{\boldsymbol{\phi}} \times \hat{\mathbf{R}} = \hat{\boldsymbol{\theta}}$
Dot product $\mathbf{A} \cdot \mathbf{B} =$	$A_x B_x + A_y B_y + A_z B_z$	$A_r B_r + A_\phi B_\phi + A_z B_z$	$A_R B_R + A_\theta B_\theta + A_\phi B_\phi$
Cross product $\mathbf{A} \times \mathbf{B} =$	$\begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$	$\begin{vmatrix} \hat{\mathbf{r}} & \hat{\boldsymbol{\phi}} & \hat{\mathbf{z}} \\ A_r & A_\phi & A_z \\ B_r & B_\phi & B_z \end{vmatrix}$	$\begin{vmatrix} \hat{\mathbf{R}} & \hat{\boldsymbol{\theta}} & \hat{\boldsymbol{\phi}} \\ A_R & A_\theta & A_\phi \\ B_R & B_\theta & B_\phi \end{vmatrix}$
Differential length $d\mathbf{l} =$	$\hat{\mathbf{x}} dx + \hat{\mathbf{y}} dy + \hat{\mathbf{z}} dz$	$\hat{\mathbf{r}} dr + \hat{\boldsymbol{\phi}} r d\phi + \hat{\mathbf{z}} dz$	$\hat{\mathbf{R}} dR + \hat{\boldsymbol{\theta}} R d\theta + \hat{\boldsymbol{\phi}} R \sin \theta d\phi$
Differential surface areas	$ds_x = \hat{\mathbf{x}} dy dz$ $ds_y = \hat{\mathbf{y}} dx dz$ $ds_z = \hat{\mathbf{z}} dx dy$	$ds_r = \hat{\mathbf{r}} r d\phi dz$ $ds_\phi = \hat{\boldsymbol{\phi}} dr dz$ $ds_z = \hat{\mathbf{z}} r dr d\phi$	$ds_R = \hat{\mathbf{R}} R^2 \sin \theta d\theta d\phi$ $ds_\theta = \hat{\boldsymbol{\theta}} R \sin \theta dR d\phi$ $ds_\phi = \hat{\boldsymbol{\phi}} R dR d\theta$
Differential volume $dV =$	$dx dy dz$	$r dr d\phi dz$	$R^2 \sin \theta dR d\theta d\phi$

Table 3-2 Coordinate transformation relations.

Transformation	Coordinate Variables	Unit Vectors	Vector Components
Cartesian to cylindrical	$r = \sqrt{x^2 + y^2}$ $\phi = \tan^{-1}(y/x)$ $z = z$	$\hat{\mathbf{r}} = \hat{\mathbf{x}} \cos \phi + \hat{\mathbf{y}} \sin \phi$ $\hat{\boldsymbol{\phi}} = -\hat{\mathbf{x}} \sin \phi + \hat{\mathbf{y}} \cos \phi$ $\hat{\mathbf{z}} = \hat{\mathbf{z}}$	$A_r = A_x \cos \phi + A_y \sin \phi$ $A_\phi = -A_x \sin \phi + A_y \cos \phi$ $A_z = A_z$
Cylindrical to Cartesian	$x = r \cos \phi$ $y = r \sin \phi$ $z = z$	$\hat{\mathbf{x}} = \hat{\mathbf{r}} \cos \phi - \hat{\boldsymbol{\phi}} \sin \phi$ $\hat{\mathbf{y}} = \hat{\mathbf{r}} \sin \phi + \hat{\boldsymbol{\phi}} \cos \phi$ $\hat{\mathbf{z}} = \hat{\mathbf{z}}$	$A_x = A_r \cos \phi - A_\phi \sin \phi$ $A_y = A_r \sin \phi + A_\phi \cos \phi$ $A_z = A_z$
Cartesian to spherical	$R = \sqrt{x^2 + y^2 + z^2}$ $\theta = \tan^{-1}[\sqrt{x^2 + y^2}/z]$ $\phi = \tan^{-1}(y/x)$	$\hat{\mathbf{R}} = \hat{\mathbf{x}} \sin \theta \cos \phi + \hat{\mathbf{y}} \sin \theta \sin \phi + \hat{\mathbf{z}} \cos \theta$ $\hat{\boldsymbol{\theta}} = \hat{\mathbf{x}} \cos \theta \cos \phi + \hat{\mathbf{y}} \cos \theta \sin \phi - \hat{\mathbf{z}} \sin \theta$ $\hat{\boldsymbol{\phi}} = -\hat{\mathbf{x}} \sin \phi + \hat{\mathbf{y}} \cos \phi$	$A_R = A_x \sin \theta \cos \phi + A_y \sin \theta \sin \phi + A_z \cos \theta$ $A_\theta = A_x \cos \theta \cos \phi + A_y \cos \theta \sin \phi - A_z \sin \theta$ $A_\phi = -A_x \sin \phi + A_y \cos \phi$
Spherical to Cartesian	$x = R \sin \theta \cos \phi$ $y = R \sin \theta \sin \phi$ $z = R \cos \theta$	$\hat{\mathbf{x}} = \hat{\mathbf{R}} \sin \theta \cos \phi + \hat{\boldsymbol{\theta}} \cos \theta \cos \phi - \hat{\boldsymbol{\phi}} \sin \phi$ $\hat{\mathbf{y}} = \hat{\mathbf{R}} \sin \theta \sin \phi + \hat{\boldsymbol{\theta}} \cos \theta \sin \phi + \hat{\boldsymbol{\phi}} \cos \phi$ $\hat{\mathbf{z}} = \hat{\mathbf{R}} \cos \theta - \hat{\boldsymbol{\theta}} \sin \theta$	$A_x = A_R \sin \theta \cos \phi + A_\theta \cos \theta \cos \phi - A_\phi \sin \phi$ $A_y = A_R \sin \theta \sin \phi + A_\theta \cos \theta \sin \phi + A_\phi \cos \phi$ $A_z = A_R \cos \theta - A_\theta \sin \theta$
Cylindrical to spherical	$R = \sqrt{r^2 + z^2}$ $\theta = \tan^{-1}(r/z)$ $\phi = \phi$	$\hat{\mathbf{R}} = \hat{\mathbf{r}} \sin \theta + \hat{\mathbf{z}} \cos \theta$ $\hat{\boldsymbol{\theta}} = \hat{\mathbf{r}} \cos \theta - \hat{\mathbf{z}} \sin \theta$ $\hat{\boldsymbol{\phi}} = \hat{\boldsymbol{\phi}}$	$A_R = A_r \sin \theta + A_z \cos \theta$ $A_\theta = A_r \cos \theta - A_z \sin \theta$ $A_\phi = A_\phi$
Spherical to cylindrical	$r = R \sin \theta$ $\phi = \phi$ $z = R \cos \theta$	$\hat{\mathbf{r}} = \hat{\mathbf{R}} \sin \theta + \hat{\boldsymbol{\theta}} \cos \theta$ $\hat{\boldsymbol{\phi}} = \hat{\boldsymbol{\phi}}$ $\hat{\mathbf{z}} = \hat{\mathbf{R}} \cos \theta - \hat{\boldsymbol{\theta}} \sin \theta$	$A_r = A_R \sin \theta + A_\theta \cos \theta$ $A_\phi = A_\phi$ $A_z = A_R \cos \theta - A_\theta \sin \theta$

GRADIENT, DIVERGENCE, CURL, & LAPLACIAN OPERATORS CARTESIAN (RECTANGULAR) COORDINATES (x, y, z)

$$\nabla V = \hat{x} \frac{\partial V}{\partial x} + \hat{y} \frac{\partial V}{\partial y} + \hat{z} \frac{\partial V}{\partial z}$$

$$\nabla \cdot \mathbf{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$$

$$\nabla \times \mathbf{A} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix} = \hat{x} \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right) + \hat{y} \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right) + \hat{z} \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right)$$

$$\nabla^2 V = \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2}$$

CYLINDRICAL COORDINATES (r, ϕ, z)

$$\nabla V = \hat{r} \frac{\partial V}{\partial r} + \hat{\phi} \frac{1}{r} \frac{\partial V}{\partial \phi} + \hat{z} \frac{\partial V}{\partial z}$$

$$\nabla \cdot \mathbf{A} = \frac{1}{r} \frac{\partial}{\partial r} (r A_r) + \frac{1}{r} \frac{\partial A_\phi}{\partial \phi} + \frac{\partial A_z}{\partial z}$$

$$\nabla \times \mathbf{A} = \frac{1}{r} \begin{vmatrix} \hat{r} & \hat{\phi} & \hat{z} \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \phi} & \frac{\partial}{\partial z} \\ A_r & r A_\phi & A_z \end{vmatrix} = \hat{r} \left(\frac{1}{r} \frac{\partial A_z}{\partial \phi} - \frac{\partial A_\phi}{\partial z} \right) + \hat{\phi} \left(\frac{\partial A_r}{\partial z} - \frac{\partial A_z}{\partial r} \right) + \hat{z} \frac{1}{r} \left[\frac{\partial}{\partial r} (r A_\phi) - \frac{\partial A_r}{\partial \phi} \right]$$

$$\nabla^2 V = \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial V}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 V}{\partial \phi^2} + \frac{\partial^2 V}{\partial z^2}$$

SPHERICAL COORDINATES (R, θ, ϕ)

$$\nabla V = \hat{R} \frac{\partial V}{\partial R} + \hat{\theta} \frac{1}{R} \frac{\partial V}{\partial \theta} + \hat{\phi} \frac{1}{R \sin \theta} \frac{\partial V}{\partial \phi}$$

$$\nabla \cdot \mathbf{A} = \frac{1}{R^2} \frac{\partial}{\partial R} (R^2 A_R) + \frac{1}{R \sin \theta} \frac{\partial}{\partial \theta} (A_\theta \sin \theta) + \frac{1}{R \sin \theta} \frac{\partial A_\phi}{\partial \phi}$$

$$\begin{aligned} \nabla \times \mathbf{A} &= \frac{1}{R^2 \sin \theta} \begin{vmatrix} \hat{R} & \hat{\theta} & \hat{\phi} \\ \frac{\partial}{\partial R} & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial \phi} \\ A_R & R A_\theta & (R \sin \theta) A_\phi \end{vmatrix} \\ &= \hat{R} \frac{1}{R \sin \theta} \left[\frac{\partial}{\partial \theta} (A_\phi \sin \theta) - \frac{\partial A_\theta}{\partial \phi} \right] + \hat{\theta} \frac{1}{R} \left[\frac{1}{\sin \theta} \frac{\partial A_R}{\partial \phi} - \frac{\partial}{\partial R} (R A_\phi) \right] + \hat{\phi} \frac{1}{R} \left[\frac{\partial}{\partial R} (R A_\theta) - \frac{\partial A_R}{\partial \theta} \right] \end{aligned}$$

$$\nabla^2 V = \frac{1}{R^2} \frac{\partial}{\partial R} \left(R^2 \frac{\partial V}{\partial R} \right) + \frac{1}{R^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial V}{\partial \theta} \right) + \frac{1}{R^2 \sin^2 \theta} \frac{\partial^2 V}{\partial \phi^2}$$

SOME USEFUL VECTOR IDENTITIES

$$\mathbf{A} \cdot \mathbf{B} = AB \cos \theta_{AB}$$

Scalar (or dot) product

$$\mathbf{A} \times \mathbf{B} = \hat{\mathbf{n}} AB \sin \theta_{AB}$$

Vector (or cross) product, $\hat{\mathbf{n}}$ normal to plane containing $\mathbf{A}$ and $\mathbf{B}$

$$\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = \mathbf{B} \cdot (\mathbf{C} \times \mathbf{A}) = \mathbf{C} \cdot (\mathbf{A} \times \mathbf{B})$$

$$\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = \mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{C}(\mathbf{A} \cdot \mathbf{B})$$

$$\nabla(U + V) = \nabla U + \nabla V$$

$$\nabla(UV) = U\nabla V + V\nabla U$$

$$\nabla \cdot (\mathbf{A} + \mathbf{B}) = \nabla \cdot \mathbf{A} + \nabla \cdot \mathbf{B}$$

$$\nabla \cdot (U\mathbf{A}) = U\nabla \cdot \mathbf{A} + \mathbf{A} \cdot \nabla U$$

$$\nabla \times (U\mathbf{A}) = U\nabla \times \mathbf{A} + \nabla U \times \mathbf{A}$$

$$\nabla \times (\mathbf{A} + \mathbf{B}) = \nabla \times \mathbf{A} + \nabla \times \mathbf{B}$$

$$\nabla \cdot (\mathbf{A} \times \mathbf{B}) = \mathbf{B} \cdot (\nabla \times \mathbf{A}) - \mathbf{A} \cdot (\nabla \times \mathbf{B})$$

$$\nabla \cdot (\nabla \times \mathbf{A}) = 0$$

$$\nabla \times \nabla V = 0$$

$$\nabla \cdot \nabla V = \nabla^2 V$$

$$\nabla \times \nabla \times \mathbf{A} = \nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A}$$

$$\int_V (\nabla \cdot \mathbf{A}) dV = \oint_S \mathbf{A} \cdot d\mathbf{s}$$

Divergence theorem (S encloses V)

$$\int_S (\nabla \times \mathbf{A}) \cdot d\mathbf{s} = \oint_C \mathbf{A} \cdot d\mathbf{l}$$

Stokes's theorem (S bounded by C)

Mathematical and Physical Models

Distance between Two Points

$$d = [(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2]^{1/2}$$

$$d = [r_2^2 + r_1^2 - 2r_1 r_2 \cos(\phi_2 - \phi_1) + (z_2 - z_1)^2]^{1/2}$$

$$d = \{R_2^2 + R_1^2 - 2R_1 R_2 [\cos \theta_2 \cos \theta_1 + \sin \theta_1 \sin \theta_2 \cos(\phi_2 - \phi_1)]\}^{1/2}$$

Coordinate Systems Table 3-1

Coordinate Transformations Table 3-2

Vector Products

$$\mathbf{A} \cdot \mathbf{B} = AB \cos \theta_{AB}$$

$$\mathbf{A} \times \mathbf{B} = \hat{\mathbf{n}} AB \sin \theta_{AB}$$

$$\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = \mathbf{B} \cdot (\mathbf{C} \times \mathbf{A}) = \mathbf{C} \cdot (\mathbf{A} \times \mathbf{B})$$

$$\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = \mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{C}(\mathbf{A} \cdot \mathbf{B})$$

Divergence Theorem

$$\int_V \nabla \cdot \mathbf{E} dV = \oint_S \mathbf{E} \cdot d\mathbf{s}$$

Vector Operators

$$\nabla T = \hat{\mathbf{x}} \frac{\partial T}{\partial x} + \hat{\mathbf{y}} \frac{\partial T}{\partial y} + \hat{\mathbf{z}} \frac{\partial T}{\partial z}$$

$$\nabla \cdot \mathbf{E} = \frac{\partial E_x}{\partial x} + \frac{\partial E_y}{\partial y} + \frac{\partial E_z}{\partial z}$$

$$\nabla \times \mathbf{B} = \hat{\mathbf{x}} \left(\frac{\partial B_z}{\partial y} - \frac{\partial B_y}{\partial z} \right) + \hat{\mathbf{y}} \left(\frac{\partial B_x}{\partial z} - \frac{\partial B_z}{\partial x} \right) + \hat{\mathbf{z}} \left(\frac{\partial B_y}{\partial x} - \frac{\partial B_x}{\partial y} \right)$$

$$\nabla^2 V = \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2}$$

(see back cover for cylindrical and spherical coordinates)

Stokes's Theorem

$$\int_S (\nabla \times \mathbf{B}) \cdot d\mathbf{s} = \oint_C \mathbf{B} \cdot d\mathbf{l}$$